SAFE & RESPONSIBLE AI SYSTEMS

1 Responsible AI (RAI) Principles

The RAI framework (Block Center / Ghani) defines the values an AI system must achieve:

- **Fairness**
- **Explainability**
- **Robustness**
- **Privacy**
- **Transparency**
- **Inclusiveness**
- **Accountability**

Assignment Q10 tests the core four: **Fairness, Explainability, Accountability, Privacy** (NOT 'AI Autonomy', NOT 'Fast Deployment').

The AI value chain (who has obligations?):

A real system (e.g. an HR hiring app) is a chain: foundation model (GPT-4/Gemini) → vendor platform (SAP) → HR → the deploying organisation. Data flows up & down. We must evaluate the **whole system, not just the AI model**, and assign obligations to the **Model Developer, the Deployer, and the User** separately.

Most orgs **procure** AI for consequential uses, so RAI matters at procurement (contrast: Office365/Google Docs co-pilots for mere productivity).

2 Key Policy Areas & the Two 2023 Executive Orders

- **AI & Democracy** — deepfakes & synthetic media: codes of conduct, detection (SynthID, C2PA)
- **Pre-deployment safety** — model transparency & auditing
- **Post-deployment safety** — a 'CERT for AI' (incident response)

Two US Executive Orders on AI: **Sept 20, 2023** and **Oct 30, 2023** (EO 14110 — Safe, Secure & Trustworthy AI). See the NTIA accountable-AI report.

Pre-deployment transparency:

- **Input Data Cards** ('nutrition labels') — high-risk/safety-critical AI must disclose data sources; auditable for IP/copyright, bias, data-poisoning (e.g. GPT-4 System Card)
- **The AI Underwriters Lab** — error tolerance varies by use (air-traffic control vs spam filter); create **risk standards (AI RMF profiles)** and auditors certify compliance — like FASB (accounting) or UL (products)

3 Guardrails: what's well-understood vs not

Challenges to operationalise RAI: documentation (**ethics by design**), **measurement** (operationalise values, codify trade-offs, set capability thresholds), **evaluation** (knowledge, retrieval, robustness, adversarial robustness, reliability, accuracy, equity), and **assurance** ('what is the GAAP for AI?').

Maturity gap (Q2): data privacy is relatively well-understood — it can be **formally bounded & guaranteed mathematically** (differential privacy). **Explainability** has no such exact bounds.

Privacy-preserving training (Q4): Differential Privacy + Federated Learning (NOT quantization or data augmentation).

4 The Systems Approach to AI Design (illustrated with Fairness)

Design for the **system**, not just the model. Three guiding questions: **what values** should the system achieve? **how** do we build it to achieve them? **how do we validate & monitor** that it keeps achieving them? — all asking: *is the system going to have the impact we want?*

The fairness design loop (exam-critical order):

Step	Action
1	**Define** desired fairness / equity goals
2	**Measure & Detect** bias ← immediately follows 'define/analyze'
3	**Understand** root causes of bias
4	**Improve** fairness of the AI systems
5	**Mitigate** the impact by adjusting interventions
6	**Monitor & Evaluate**

Assignment Q12: after **analyzing root causes of bias**, the next action in the framework is to **measure & detect the bias**; the overall goal is to **improve the fairness of AI systems**.

5 Measuring Fairness & the Fairness Tree

Many fairness measures exist (group vs individual; output vs **process** fairness). There is an inherent **accuracy-vs-disparity trade-off**: the ideal model is high precision@k *and* low disparity, but usually improving fairness costs some accuracy.

Fairness Tree (Ghani et al.): pick the right metric from the situation — is the intervention **punitive vs assistive**, and which group matters? **Punitive intervention + predictive equity among those intervened on** → **FDR Parity** (False Discovery Rate parity). Q11 answer = **FDR Parity**.

6 Autonomy, Copyright, Energy & Adversarial Attacks

Autonomy & human agency (Q3): as AI becomes autonomous agents, **individual human agency decreases as decision-making is delegated** to the AI.
Copyright inside a model (Q6): to remove it, the **model may need retraining or specialized unlearning** (ties to Week 3).
Green AI (Q8): cut energy/carbon by **moving inference to CPUs/Edge devices instead of GPUs** for simple queries.

Suffix / adversarial attacks (Fredrikson et al.): chatbots are 'aligned' to a policy in training and follow it normally, **but fail when probed adversarially**. A **universal suffix attack** found on open-source LLMs can be pasted into public chatbots to **wrest control of the AI component, bypassing guardrails**.
Back Translation Defense (Q9): take an unmoderated output, **reverse-infer the prompt that generated it, and run that prompt through safety filters**.
Safety budget (Q7): ~**10%** of development budget on safety & responsible-AI research.

7 Post-Deployment: the AI Emergency Response Team

Even with pre-deployment diligence, failures & vulnerabilities appear after launch. Propose an **AI Lead Emergency Response Team (ALRT)** — catalog failures/incidents, do forensic analysis, risk-tier them, and share data upstream to developers & downstream to deployers. Modelled on CMU's **CERT** (Cyber Emergency Response Team).

Vertical vs Horizontal regulation (Q1): vertical = apply existing **sector-specific** laws to AI; **horizontal** = **broad new risk-based laws across all sectors**. **Co-regulation (Q5):** industry sets policies that are **formally recognized & enforced by government**. **Upstream > downstream (Q11):** regulate developers, since **users face high market pressure to adopt AI**.